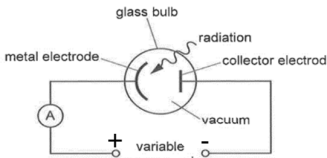
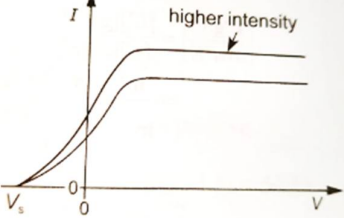


	ends X and Y is zero.
7(a)(i)	Photoelectric effect
7(a)(ii)	Diffraction in Young's double slit experiment
7(b)(i)	 The diagram illustrates a vacuum tube setup for the photoelectric effect. It consists of a glass bulb containing a vacuum. Inside the bulb, there are two electrodes: a curved metal electrode (cathode) and a straight collector electrode (anode). Radiation is incident on the metal electrode, causing the emission of photoelectrons. A circuit is connected to the electrodes, including an ammeter (A) and a variable power supply with positive (+) and negative (-) terminals.

7(b)(ii)	$hf = \Phi + KE_{\max}$ $KE_{\max} = hf - \Phi$ $= 12.4 - 4.2$ $= 8.2 \text{ eV}$ $eV_s = 8.2(1.6 \times 10^{-19})$ $V_s = \frac{8.2(1.6 \times 10^{-19})}{1.6 \times 10^{-19}} = 8.2 \text{ V}$
7(b)(iii)	Electromagnetic (EM) radiation arrives at the metal electrode as discrete packets of energy known as photons, each with energy $E = hf$, where f is the frequency of EM radiation. Each photon interacts with an electron in the metal surface on a one-to-one basis and one photon can only transfer energy to one photoelectron. When the intensity of EM radiation is increased, more photons arrive at the metal electrode per unit time. However the energy of each photon is the same since the frequency of the source does not change and the metal used does not change. Thus the maximum kinetic energy of the emitted photoelectrons remains the same, and the same stopping potential is required to prevent current flow.
7(c)(i) and 7(c)(ii)	
8(a)	From Fig. 8.1, the period of the pulsar = 0.60 s